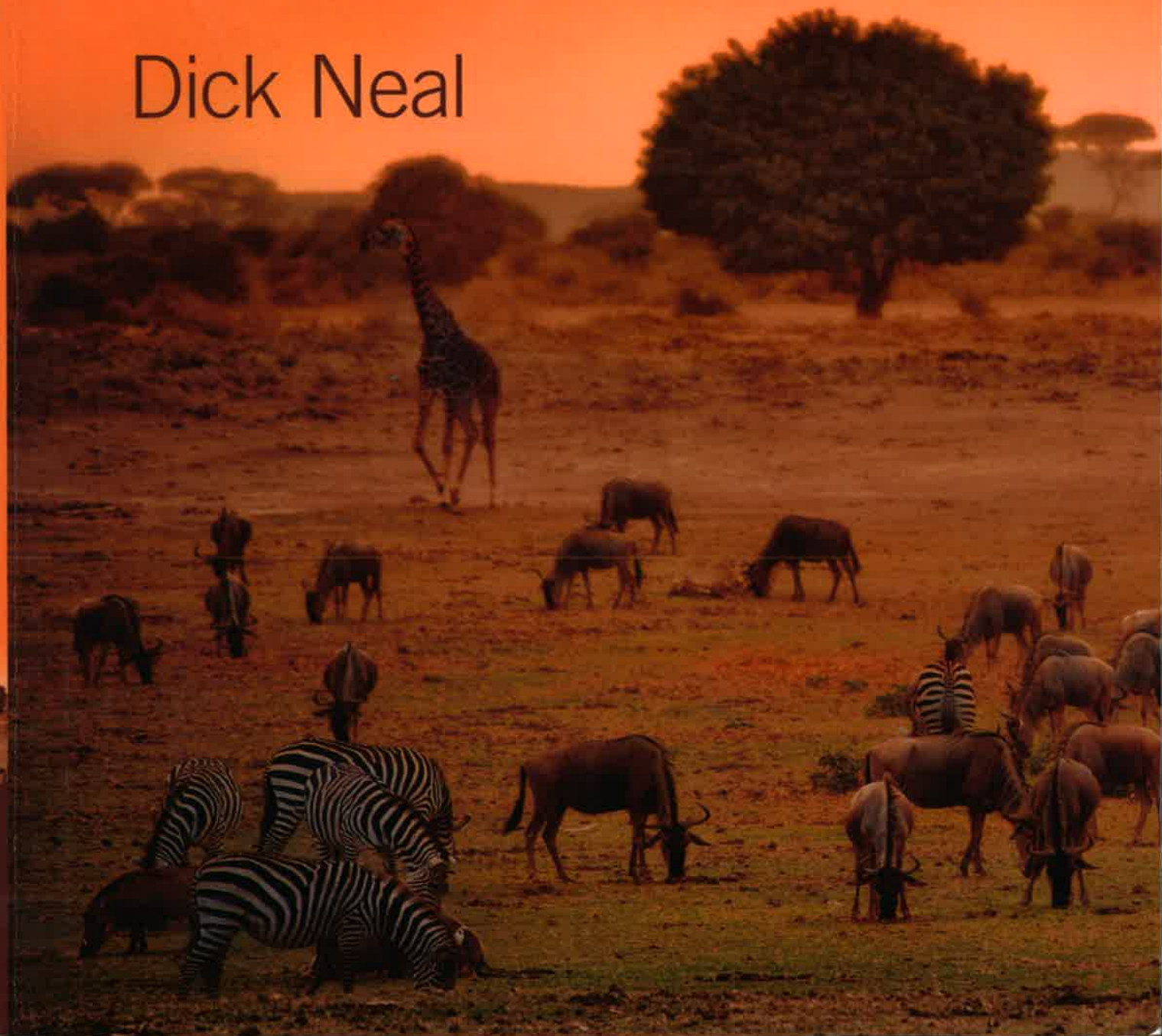


Introduction to Population Biology

Second Edition

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In Chapter 4, we learned that populations must have the potential to increase exponentially in size, but that such growth is unsustainable in the long term because resources are finite. As populations increase in size, the competition for resources increases. Darwin recognized that this competition was universal in organisms and called it the struggle for existence. The struggle for resources involves both intra- and interspecific competition (see Chapter 18), but in this chapter we will focus on developing models that describe how population growth may be influenced by population density through the effects of intraspecific competition for resources.

As populations increase in density, the resources needed to sustain them become limiting. For example, barnacles may cover the entire surface of a rock until there is no more space available for further growth of the population. Similarly, cavity-nesting birds may have the size of the breeding population limited by the availability of suitable holes in trees. In some cases, the population may far exceed what can be sustained by the environment, and here we may observe a reduction in population size. For example, many plants release a prolific number of seeds and, as self-thinning occurs, we can observe intraspecific competition as the seedlings increase in size. In one experiment on fleabane (*Conyza canadensis*), the density of the seedling population declined from 124 400 per m² to 1060 adult plants per m² (Harper 1977). Thus, growth will depend on the density of the population in relation to the availability of resources. The basic premise is that the realized growth rate will decline as population density increases, and if the density exceeds what can be sustained by the available resources, the growth rate will be negative.

5.1 THE LOGISTIC GROWTH MODEL

The logistic growth equation was first formulated by Verhulst (1838) and then rediscovered by Pearl and Reed (1920). Pearl promoted its use as a universal law of population growth, but this proved to be controversial (Kingsland 1982, 1991). The criticism is well founded because the model is based on some highly restrictive assumptions, and more realistic models are available (e.g. Law *et al.* 2003). My reasons for considering the model are threefold: (1) the model has had a profound effect on the development of ideas in population ecology that we need to understand; (2) the changes in gene frequencies as a result of selection are logistic in some cases (see Chapter 14); and (3) the logistic growth equation does provide a reasonable fit to the growth of some natural populations (see Section 5.5), contrary to the assessment by Hall (1988). The model can be developed in many ways. The following attempts to be a student-friendly version.

Imagine a population being introduced into an area that has sufficient resources to sustain a stable population density of K individuals, called the carrying capacity of the population. When the population attains a size of N individuals, the population is using N/K of the resources leaving $1 - N/K$ of the resources available for further growth of the population. If we assume that all individuals are identical, each individual uses $1/K$ of the available resources and reduces the per capita growth rate of the population by $1/K$.

Thus, the per capita growth rate, r , is a function of density and declines as population density increases. In populations where K is large, the value of r will approximate r_m when the population density (N) is very small because the density effects are minimal and growth is approximately exponential. However, as the density of the population increases, the value of r declines steadily until it equals zero at the carrying capacity and the population stops growing. We can express this relationship for the per capita growth rate as follows:

$$r = r_m \left(1 - \frac{N}{K} \right) \quad (\text{Eqn 5.1})$$

This equation shows that the growth rate per capita (r) steadily declines as the population size increases, and if N exceeds K , it becomes negative and so the population density will decline. Consequently, K represents an equilibrium density for the population.

The change in density of the population is predicted by the equation $\delta N / \delta t = rN$. If we substitute the right-hand side of Eqn 5.1 for r , we obtain the familiar logistic growth equation:

$$\frac{\delta N}{\delta t} = r_m N \left(1 - \frac{N}{K} \right) \quad (\text{Eqn 5.2})$$

This is frequently presented in two other equivalent forms:

$$\frac{\partial N}{\partial t} = r_m N - \left(\frac{r_m N^2}{K} \right) \quad (\text{Eqn 5.2a})$$

$$\frac{\delta N}{\delta t} = r_m N \left(\frac{K - N}{K} \right) \quad (\text{Eqn 5.2b})$$

One interpretation of Eqn 5.2a is that the rate of increase of the population ($\delta N / \delta t$) is equal to the biotic potential, i.e. the potential for exponential growth ($r_m N$), minus the resistance to growth that is created by the population itself, i.e. density-dependent effects ($r_m N^2 / K$). This latter term can be considered to be a measure of intraspecific competition and is one component of what Darwin termed the 'struggle for existence'.

To express population density as a function of time, Eqn 5.2 is integrated following the rules of integral calculus to derive the following complex equation (see Otto and Day 2007):

$$N_t = \frac{e^{r_m t}}{1 - \frac{N_0}{K} + \frac{e^{r_m t} N_0}{K}}$$

This simplifies to:

$$N_t = \frac{K}{1 + \left(\frac{K - N_0}{N_0} \right) e^{-r_m t}} \quad (\text{Eqn 5.3})$$

Thus, N_t is related to the starting population size (N_0), the carrying capacity (K) and the intrinsic rate of natural increase (r_m) in a complex way.

If we wish to calculate r_m from a logistic growth curve, when we know the population density at three points (N_0 , N_t and K), we can rearrange and simplify Eqn 5.3 to:

$$-r_m t = \ln \left[\frac{N_0(K - N_t)}{N_t(K - N_0)} \right] \quad (\text{Eqn 5.4})$$

The following are examples of how these equations may be applied.

EXAMPLE 5.1

The growth of a laboratory culture of Paramecium was accurately predicted by the logistic growth equation. If the equilibrium density (K) is 400 individuals per ml, and the intrinsic rate of natural increase (r_m) is 0.7 per day, what is the predicted density of individuals after 10 days in a culture started with five individuals per ml?

Use Eqn 5.3 to solve this. The value of $e^{-0.7 \times 10} = 0.000912$, and this is multiplied by $400/5 - 1$ (=79) to obtain 0.072. Add one to this value (= 1.072), and then divide the sum into 400 to obtain the answer of 373 individuals per ml (rounded to the nearest integer).

EXAMPLE 5.2

A population of songbirds has an equilibrium density of 31 breeding pairs per hectare. A population was introduced into a new area at a density of one breeding pair per hectare and reached a density of 12 breeding pairs per hectare after 10 years. (a) What is the intrinsic rate of natural increase (r_m) assuming that the population is growing logistically? (b) What is the realized rate of increase per capita when there are 12 breeding pairs in the population?

- (a) Use Eqn 5.4 and set $t = 10$, $K = 31$, $N_t = 12$ and $N_0 = 1$. The answer is 0.294 per year.
 (b) Use Eqn 5.1 to calculate this, where $r_m = 0.294$ per year, $N = 12$ and $K = 31$. The answer is approximately 0.180 per year. Note that r does not appear in Eqns 5.2–5.4, because these equations automatically calculate the realized r values from the r_m , N and K values.

5.2 SIMULATING LOGISTIC GROWTH

You may simulate the predictions of Eqns 5.1–5.3 by going to Chapter 5 of the Simulations website and completing the exercises. Our simulations provide graphs that are similar to Figure 5.1.

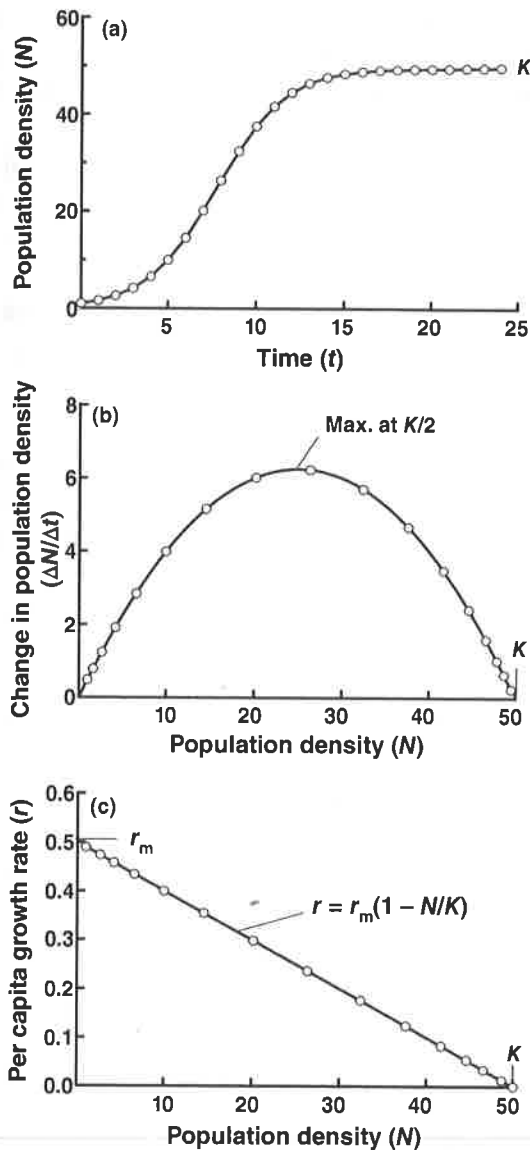


Figure 5.1 (a) Logistic growth of a population with a carrying capacity (K) of 50, an r_m value of 0.5, starting from N_0 of 1 individual. (b) The change in population density ($\Delta N/\Delta t$) as a function of density (N) for the population in Figure 5.1a. (c) Per capita growth rates (r) as a function of density (N) for a population undergoing logistic growth.

The logistic model of population growth predicts that populations attain a stable carrying capacity. The form of growth is S-shaped for populations starting at a density below that of the carrying capacity (Figure 5.1a), and so it is sometimes called sigmoidal growth. The precise shape of the curve depends on the starting density (N_0) and the carrying capacity (K), and the steepness of the curve is directly proportional to the value of the intrinsic rate of increase (r_m). Population densities never overshoot the carrying capacity, and so the growth curves have a smooth shape. This indicates a perfect adjustment of the growth rate, r , as the density changes.

Simulations show that populations starting at densities above the carrying capacity approach the carrying capacity more rapidly than populations starting at densities below the carrying capacity. This is because the inhibition to population growth (term $r_m N^2/K$ in Eqn 5.2a) is related to the square of the population density.

An examination of the S-shaped growth curve suggests that the population grows at its fastest rate at intermediate densities. This observation is confirmed when the population growth rate is plotted as a function of population density (Figure 5.1b). The maximum increase in density always occurs at half the carrying capacity (i.e. $K/2$).

The growth rate per capita, or r_m , remains constant in exponential growth (Chapter 4). In contrast, the growth rate per capita (r) in the logistic growth model declines linearly as the density increases (Figure 5.1c). When the density is close to zero, r is approximately equal to r_m because there are no effects of density. The value of r declines to zero when the density reaches the carrying capacity, and at densities above the carrying capacity, r is negative. Assuming no immigration or emigration, the population adjusts the value of r in relation to density by altering the birth and death rates, and a stable equilibrium ($N = K$) is reached at a density where the birth rate is equal to the death rate.

The model has many unrealistic assumptions. It assumes that all individuals are identical, but in reality they may vary in size, age, sex and genotype. These factors affect birth and death rates, and the use of resources, and so we cannot expect r_m and K to be constants. The model also assumes that individuals adjust their birth and death rates (i.e. r) instantaneously as the population changes in size, but in reality there will be time lags to any such response (see Section 5.3). Finally, it assumes that the environment is constant, but environments change over the course of time and this is another reason why we cannot expect r_m and K to be constants.

Let us relax some of the restrictive assumptions of the model to see how the form of growth may change.

5.3 TIME LAGS

Most populations have time lags in the way that they adjust their birth and death rates in relation to population density. For example, many species lay eggs that hatch independently of the parent, and so the birth rate cannot be adjusted if the population density changes between the laying and hatching of the eggs. In this case, the birth rate is related to the density at the time of egg deposition, not the time of hatching, and the time lag will correspond to the length of the incubation period. Similarly, when the young are born, they are usually much smaller than adults. As they grow in size, they require more resources, and the death rate may adjust as a consequence. In this case, the time lag will be related in some way to the developmental period of the young.

There are various models to simulate time lags in logistic growth, but we will consider only one of them. The discrete version of the logistic model describes population growth by the following equation:

$$N_{t+1} = N_t + r_m N_t \left(1 - \frac{N_t}{K} \right) \quad (\text{Eqn 5.5})$$

The population size at time $t + 1$ depends on the population size at time t plus the change in population size predicted by Eqn 5.2 at time t . The growth rate is not adjusted as the population increases from N_t to N_{t+1} because there is a built-in time lag of one time step. As the time lag is a constant, the response of the model depends solely on r_m .

5.3.1 Simulating Time Lags: a Discrete Version of the Logistic Growth Model

The model's behaviour is surprisingly complex (see Simulations website). At low values of r_m , it behaves like the simple logistic model and smoothly approaches the carrying capacity in the familiar S-shaped pattern of growth (Figure 5.2a). When r_m attains a value of about 1.1, the population first overshoots and then undershoots the carrying capacity in a series of damped oscillations. These oscillations are barely evident at first but become more noticeable as r_m becomes larger (Figure 5.2b). When r_m is greater than 2.0, the population begins to oscillate about the carrying capacity in a stable two-point cycle that rapidly becomes more and more complex as r_m increases from 2.449 to 2.57 (Figure 5.2c). At values above 2.57, the population fluctuates around the carrying capacity in a chaotic manner (Figure 5.2d), and when r_m is a little greater than 3, the oscillations become more extreme and lead to extinction of the population (see Simulations website).

The growth of populations starting at similar, but not identical, densities is almost identical at low values of r_m (Figure 5.2a–c), but once the fluctuations become chaotic, the two populations diverge from one another over time (Figure 5.2d). Also note that when the population fluctuates around the carrying capacity (K), the average population size is less than K , because populations above the carrying capacity decline more rapidly in size than populations below the carrying capacity increase in size.

What does all this mean? For populations that grow in a series of steps, like many annual insects and plants, their form of growth may not appear to be logistic, even when their birth and death rates are density dependent. Their form of growth depends on the value of r_m , which is often high, so we might expect their densities to be chaotic from one year to the next. However, there are limits to the size of these fluctuations, which may allow us to detect density dependence.

5.4 VARYING THE CARRYING CAPACITY

The carrying capacity is not constant because resources vary both seasonally and from year to year. From our analysis of time lags, we can obtain an understanding of how populations will respond, because there will be an inevitable lag in the population's response to changes in the carrying capacity. The population response depends on its intrinsic rate of increase (r_m).

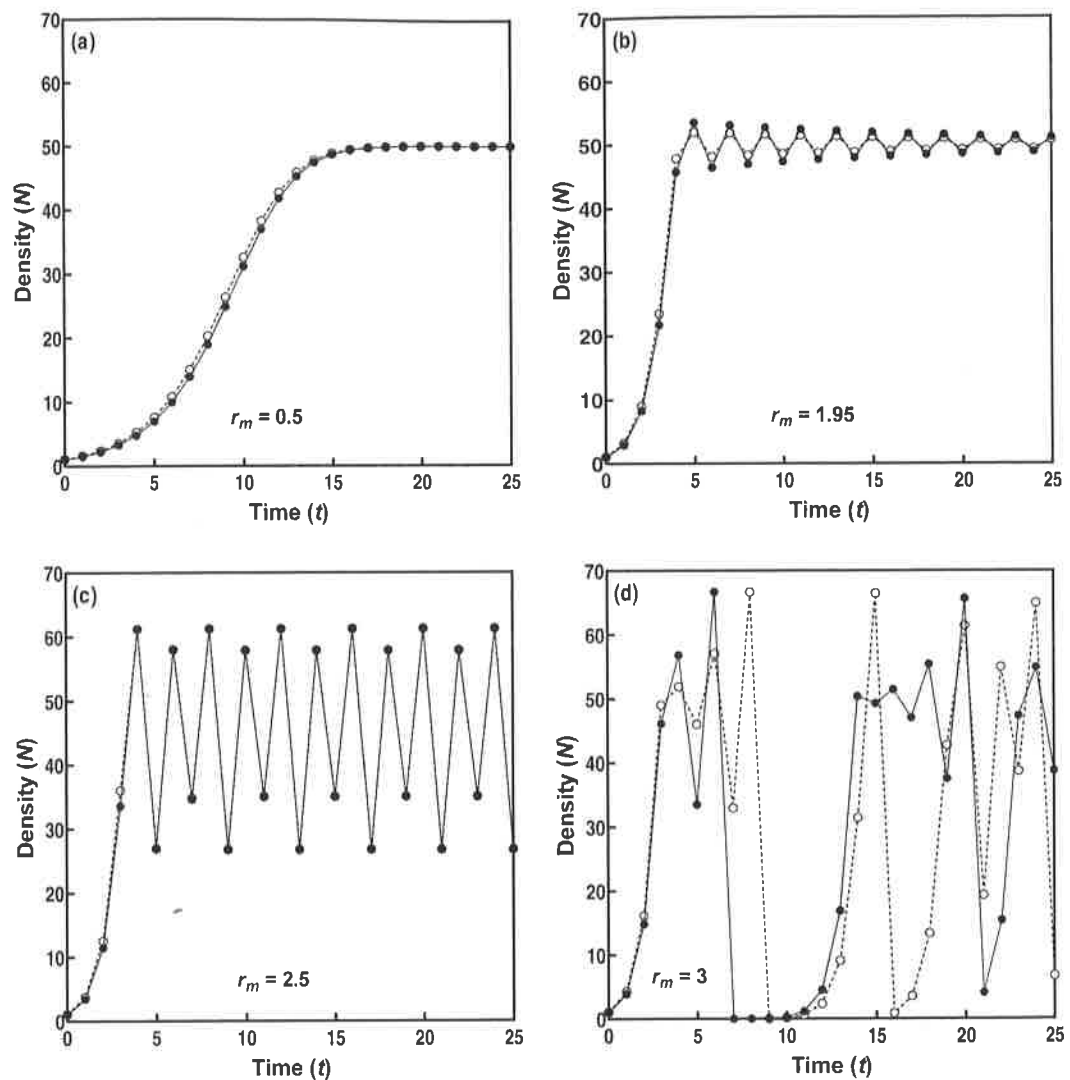


Figure 5.2 Behaviour of the discrete logistic growth model for a population with a carrying capacity (K) of 50 and different values of r_m . Solid and dashed lines indicate populations starting at 1 and 1.1 individuals, respectively. (a) $r_m = 0.5$; (b) $r_m = 1.95$; (c) $r_m = 2.5$; (d) $r_m = 3$. See text for discussion.

Populations with high r_m values, such as many species of small mammals and insects, have a short time lag and so will tend to track the fluctuations in K . In contrast, populations with low r_m values, such as large mammals, have longer time lags and so react more slowly. They will vary less than the variation in carrying capacity, and will tend to persist at a density that is lower than the overall average of K . Remember that fluctuations in density are asymmetrical about K , and that the mean density is below K (Figs. 5.2c, d). Excellent introductions to this topic are provided by May (1976) and Gotelli (2008).

5.5 EXAMPLES OF LOGISTIC GROWTH

5.5.1 Yeast

The growth of laboratory cultures of unicellular organisms is often logistic because we can maintain a constant carrying capacity and any time lags are likely to be short. Yeast is a good example (Figure 5.3), where it may be seen that the logistic growth curve provides an excellent fit to these data. Yeast use sugars for their energy supply, producing by-products such as ethyl alcohol and carbon dioxide that build up in the medium. At some point, they stop growing and enter a resting phase. The Simulations website demonstrates how you can fit logistic growth curves for this type of data.

5.5.2 Ponderosa Pine

Lesser and Jackson (2012) studied four isolated populations of ponderosa pine (*Pinus ponderosa*), growing in 10–173-ha areas in the Bighorn Basin in Wyoming. Between 84 and 90 per cent of each population was aged, and the establishment and growth of each population was reconstructed from its age structure. This was possible because the area is arid and dead wood remains for centuries. The only deaths in the four populations had occurred in the last 5–10 years (many still had needles): 17 at Cottonwood Creek and seven at Anchor Dam. All four populations exhibited logistic-like growth (Figure 5.4) following initial colonization episodes between the years of 1531 at Cottonwood Creek and 1655 at Anchor Dam. Following the initial colonization, population growth was slow because self-fertilization is rare in this species and so growth was probably influenced more by rare episodes of immigration. An extreme example of this is seen at Grass Creek where, following the establishment of the first individual in 1605, it took over 130 years before a second individual became established. This initial phase of slow growth was followed by a long period of faster growth, from the mid-1700s until the early 1900s in three of the populations, and from 1800 to 1950 in the Anchor Dam population. The population growth then slowed, sometimes quite abruptly, as shown by the Anchor Dam

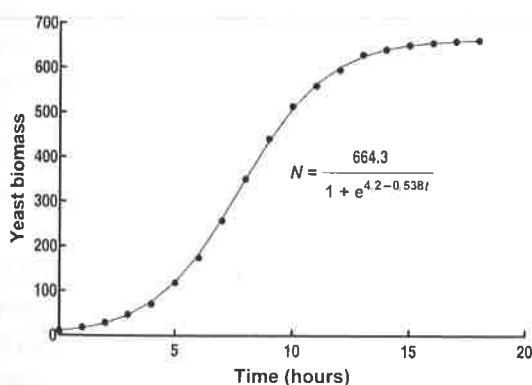


Figure 5.3 Growth of a population of yeast cells showing the fit of a logistic growth curve. See Simulations website for details. (Data from Carlson 1913, cited in Pearl 1927.)

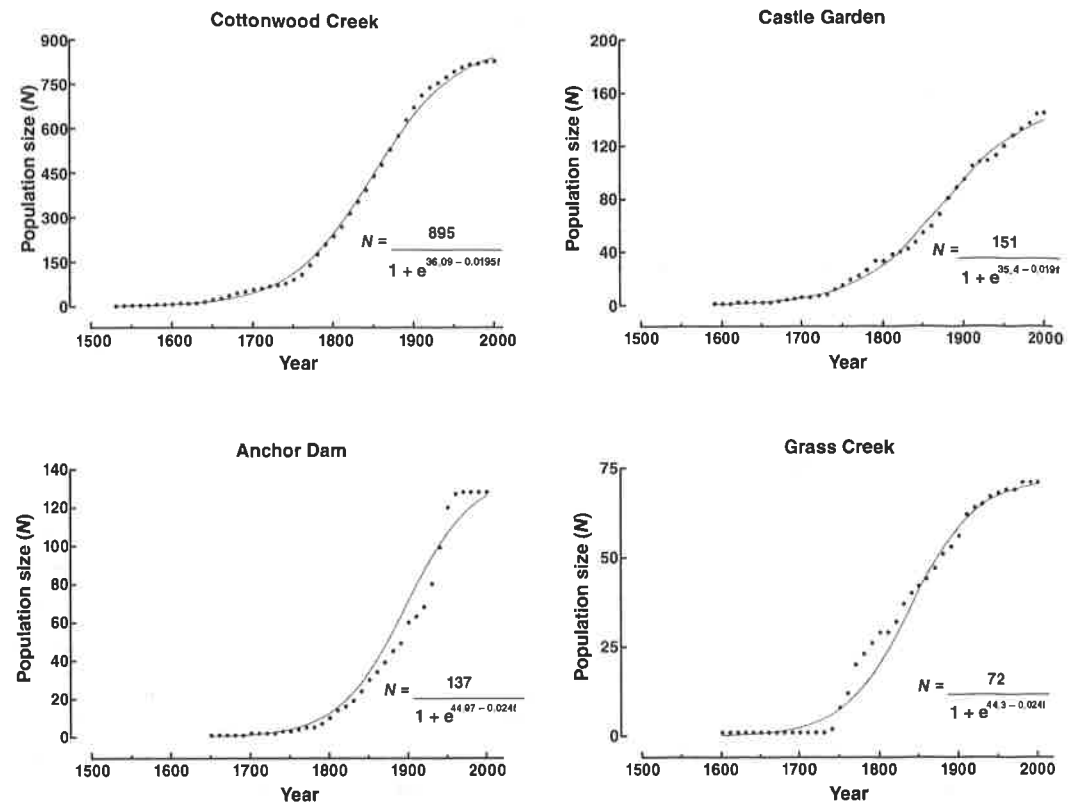


Figure 5.4 Approximate logistic growth of four discrete populations of ponderosa pine (*Pinus ponderosa*) in Wyoming, USA, following their establishment between 1531 and 1655. See website simulations for details. (Data from Lesser and Jackson 2012.)

population, and there has been little or no increase since 2000. Although the four populations appear to be reaching an asymptote, this is not a result of density effects because the densities of the four populations range from 0.6 to 12.8 individuals per hectare. More probably, the reduction of recruitment is related to increased grazing of seedlings by cattle or the effects of climate (Lesser and Jackson 2012). Consequently, the fact that the logistic model provides a reasonable fit to the data does not imply that the model provides an explanation for the growth form of these populations.

5.5.3 *Paramecium*

The growth of *Paramecium caudatum* does not describe such a near-perfect logistic growth form as yeast, although the pattern of growth is S-shaped (Figure 5.5). Its growth form suggests that the population may be oscillating about the carrying capacity. There are two possible reasons for this. There may be a time lag operating, although the value of r_m does not suggest

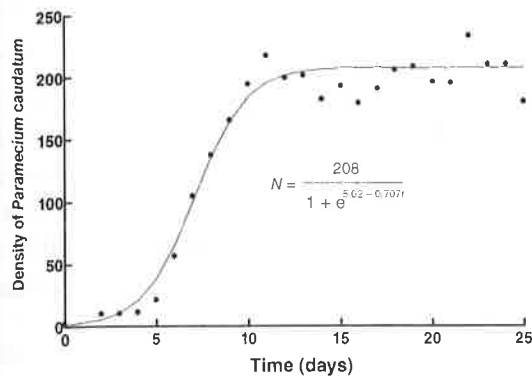


Figure 5.5 Growth of *Paramecium caudatum* in laboratory culture showing the fit of the logistic growth curve. (Data from Gause 1934.)

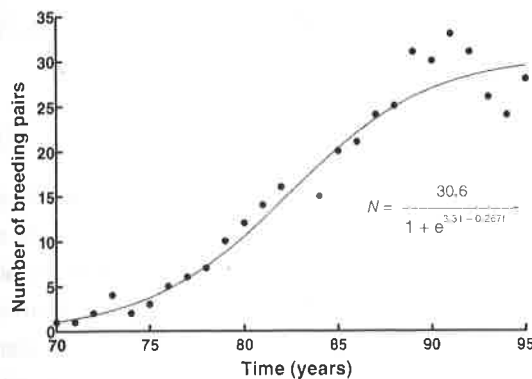


Figure 5.6 Increase in the number of breeding pairs of merlins in Saskatoon, Canada, from 1970 to 1995, with a fitted logistic growth curve (see website). (Data from Oliphant and Haug 1985 and Lieske 1997.)

that this would result in sustained oscillations (see Section 5.3.1). Alternatively, the population may be responding to periodic fluctuations in the environment (i.e. a variable K), such as the addition of food at fixed intervals. The apparent oscillation around the carrying capacity requires a modification of how we fit the logistic growth curve to these data (see Simulations website).

5.5.4 Merlins

In Chapter 4, we saw that the merlin population in Saskatoon, Canada, increased from one to 16 breeding pairs during the period 1970–82. If the population had continued to increase exponentially at the same rate, there would have been 365 breeding pairs in 1995. In fact, only 28 pairs were recorded, which is not surprising in view of the limitations of suitable nest sites and food availability.

The number of breeding pairs recorded over the 26-year period exhibited approximately logistic growth (Figure 5.6). The carrying capacity is not constant, however, because the availability of nest sites fluctuates. Most nests are located in mature spruce trees and their density changes as landowners cut down and replace old trees, and as new trees mature, particularly in newer neighbourhoods.

5.6 MAXIMUM SUSTAINABLE YIELD

The logistic growth model has been used to calculate the maximum sustained yield (MSY) to harvest various fish and game populations (Caughley 1977). The model is based on simple logic: if population growth conforms to the logistic growth curve, the largest increase in numbers occurs at half the carrying capacity (Figure 5.1b) and at this density the instantaneous growth rate, r , is equal to $r_m/2$ (Figure 5.1c). Therefore, to maximize the harvest we adjust the population to half its carrying capacity ($K/2$) and then harvest it at half its intrinsic rate of natural increase ($r_m/2$) to obtain:

$$\text{Instantaneous MSY} = \frac{K}{2} \times \frac{r_m}{2} = \frac{r_m K}{4} \quad (\text{Eqn 5.6})$$

The instantaneous MSY is appropriate for populations that grow and are harvested continuously. However, many populations breed or are harvested only at one time of the year. For these populations, the finite harvest rate is $1 - e^{-r_m/2}$, not $r_m/2$, so our yield is:

$$\text{Finite MSY} = \frac{K}{2} \times (1 - e^{-r_m/2}) \quad (\text{Eqn 5.7})$$

Go to the Simulations website (Chapter 5, see section on 'Calculating Maximum Sustainable Yields (MSY)') for more details about how to apply this method. The issue of estimating the carrying capacity and rate of increase for populations that have already been harvested is also considered.

The strategy can be extremely risky. The carrying capacity may fluctuate, and estimations of K and r_m are not without error; in addition, any errors accumulate so that MSY estimates may not be reliable (see Simulations website). Populations need to be monitored very carefully, and if the carrying capacity is variable, it is very easy to overfish (i.e. drive the population below $K/2$). Obviously, one should immediately reduce the harvest, but there are often political pressures to delay doing this. Once a fishery has collapsed, its recovery is by no means assured, even if the fishing levels are reduced to zero.

5.7 Summary and Conclusions

The logistic growth curve is one mathematical model that describes Darwin's 'struggle for existence' when the struggle involves only intraspecific competition for resources. The model assumes that the population consists of identical individuals that are growing in a constant environment of defined limits or size. As the population increases, intraspecific competition for resources also increases, and the individuals adjust their growth rates accordingly as they utilize the fixed resources of the environment. The model describes an S-shaped form of growth to a stable carrying capacity, K , which depends on the characteristics of the population and the number of resources available in the environment, while the steepness of the curve depends on the per capita rate of increase, r .

The restrictive assumptions of the model may be relaxed to analyse how factors such as time lags and a variable environment affect the form of population growth. These analyses show that time lags have surprisingly little effect on the form of population growth for populations with low r values. As r increases to higher values, population densities oscillate more and more, and may become unpredictable (or chaotic) and eventually are not sustainable in the long term. The way populations respond to environmental variation (affecting K) is also influenced by r . Populations with high r values track the changes in environment, whereas populations with low r values tend to average the environmental fluctuations over time.

The basic model has great heuristic value in spite of its restrictive assumptions because it can be used as a basis to investigate the pattern of population growth. Departures from the idealized S-shaped form of growth can be analysed to determine whether they are related to internal factors, such as time lags, or external factors, such as a variable environment.

5.8 Problems

The summary of equations (Box 5.1) should help you with the following problems. You will also need to remember the equations relating to exponential growth (Box 4.1).

1. Compare the relationship between (a) the rate of increase per capita (r) and density (N), and (b) the population rate of increase ($\delta N/\delta t$) and density (N), for the exponential and logistic growth models.
2. (a) You set up a colony of worms to sell to local fishermen. Starting with only five individuals, you are delighted to find that they have increased to 1044 individuals after 1 month (28 days). What is the value of r_m per day assuming exponential growth?
(b) Using the information in part (a), calculate the expected size of the population after 15 weeks (from the original starting time), assuming a constant rate of increase per capita (r_m).
(c) You have dreams of becoming a millionaire, but your hopes are dashed when you discover that there are only 2500 worms after 15 weeks and their numbers stay approximately constant thereafter. Calculate the value of r_m , assuming logistic growth.

- (d) Why are the values of r_m different in parts (a) and (c)?
- (e) What is the maximum sustainable daily harvest for this population if you assume logistic growth?
3. A population takes 10 days to double in size from 20 to 40 individuals. How long will it take to double in size again to 80 individuals if (a) it grows exponentially, or (b) it grows logistically and $K = 100$?
4. (a) The growth of a microbial population was found to be accurately predicted by the logistic growth equation. If the equilibrium density of the cells (K) is 5.0×10^6 cells per ml, what is the predicted density after 3 h in a culture that was started with a density of 2×10^3 cells per ml if the intrinsic rate of natural increase (r_m) is 0.29 per hour?
- (b) What is the maximum sustainable yield per hour of this population?
- (c) How long will it take to start harvesting at the maximum sustainable rate starting from the density given in part (a)?

BOX 5.1 SUMMARY OF EQUATIONS FOR LOGISTIC GROWTH

$$r = r_m \left(1 - \frac{N}{K} \right) \quad (\text{Eqn 5.1})$$

$$\frac{\partial N}{\partial t} = r_m N \left(1 - \frac{N}{K} \right) \quad (\text{Eqn 5.2})$$

$$\frac{\partial N}{\partial t} = r_m N - \left(\frac{r_m N^2}{K} \right) \quad (\text{Eqn 5.2a})$$

$$\frac{\partial N}{\partial t} = r_m N \left(\frac{K - N}{K} \right) \quad (\text{Eqn 5.2b})$$

$$N_t = \frac{K}{1 + \left(\frac{K - N_0}{N_0} \right) e^{-r_m t}} \quad (\text{Eqn 5.3})$$

$$-r_m t = \ln \left[\frac{N_0(K - N_t)}{N_t(K - N_0)} \right] \quad (\text{Eqn 5.4})$$